

A Provably One-Parameter Theory of Gravity and the Dark Sector: the de Sitter–MOND Coefficient $\kappa = \frac{1}{2}$ Is Unforceable Geometry

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Abstract

The de Sitter–MOND framework writes the galactic acceleration scale as $a_0 = c^2 \sqrt{\Lambda/32\pi} = \kappa c \sqrt{G\rho_{\text{DE}}}$, and the *form* $a_0 \propto c^2 \sqrt{\Lambda}$ together with the gravitational kernel $\sqrt{8\pi/3}$ (including the half-integer power of π) are forced. We ask whether the single residual coefficient $\kappa = \frac{1}{2}$ is forced or free, and prove a sharp result: **κ is the framework’s lone free number, and it is provably unforceable** by the three standard consistency conditions — ghost-freedom, unitarity, and a holographic / degree-of-freedom bound — each failing for a structurally distinct, machine-verified reason. We then identify what κ *is*: the framework coefficient $\sqrt{2/Z} = (3/8\pi)^{1/4} = 0.58779$ (with $Z = \sqrt{32\pi/3}$) is **pure geometry**, carrying no degree-of-freedom, thermal, or entropy content, and it equals exactly the **single-degree-of-freedom** ($g_* = 1$) **limit of the Cohen–Kaplan–Nelson (CKN) UV–IR bound** — the geometric identity $3/8\pi = (\text{Schwarzschild } \frac{1}{2})/(\text{ball } \frac{4\pi}{3})$. An honest count of the Standard Model’s degrees of freedom ($g_* = 106.75$) gives a CKN coefficient of $\sim 0.18\text{--}0.41$, never the framework value, so κ is **geometric, not microscopic**. The framework is therefore a *provably* one-parameter theory of gravity and the dark sector — one knob, irreducible and geometric — versus Λ CDM’s six parameters plus an undetected particle. We state plainly what this does and does not establish: it fixes the parameter economy and the irreducibility of the lone number; it does **not** derive a_0 ’s value, make the sector zero-parameter, or confirm the framework against Λ CDM. This is a structural theorem about an effective theory at a frontier — not a theory of everything yet, as frustrating as it may be.

1. The question

A modified-dynamics account of galaxy rotation organizes everything around one acceleration, $a_0 \approx 1.2 \times 10^{-10} \text{ m s}^{-2}$. The de Sitter–MOND framework proposes that this scale is set by the cosmological constant,

$$a_0 = c^2 \sqrt{\frac{\Lambda}{32\pi}} = \frac{c}{2} \sqrt{G\rho_{\text{DE}}} = \frac{cH_\Lambda}{Z}, \quad Z = \sqrt{\frac{32\pi}{3}} = 5.789,$$

evaluated on the dark-energy density alone, giving $a_0 = 9.36 \times 10^{-11} \text{ m s}^{-2}$. Prior work establishes two facts we take as given. First, the *form* $a_0 \propto c^2 \sqrt{\Lambda}$ is over-determined — multiple independent mechanisms force the de Sitter–Unruh interpolation and the $\sqrt{\Lambda}$ scaling. Second, the **gravitational kernel** $\sqrt{8\pi/3}$, including the $\sqrt{\pi}$ that no curvature-free thermal route produces, is forced by the

Einstein density normalization. What remains is a single dimensionless coefficient. Writing

$$a_0 = \kappa c \sqrt{G \rho_{\text{DE}}}, \quad \rho_{\text{DE}} = \frac{\Lambda c^2}{8\pi G},$$

the framework's value corresponds to $\kappa = \frac{1}{2}$: solving $a_0 = \kappa c \sqrt{G \rho_{\text{DE}}} = c^2 \sqrt{\Lambda/32\pi}$ gives $\kappa = \frac{1}{2}$ exactly. The question of this paper is sharp: **is κ forced, or is it the theory's one free number?**

2. The scale–fraction split

κ is the multiplicative coefficient *outside* the root, while the $\sqrt{8\pi}$ that fixes its value lives *inside* the Einstein density $\rho_{\text{DE}} = \Lambda c^2/8\pi G$. Any condition that lives on the de Sitter–Unruh temperature spectrum $W(a) = \sqrt{a^2 + (cH_\Lambda)^2}$ depends only on the dimensionless ratio a/cH_Λ — the *scale*, the inside-root structure Z — and can never reach the *outside fraction* κ , because the spectrum never sees the 8π density normalization. Symbolically, $a_0 = \kappa c^2 \sqrt{\Lambda}/\sqrt{8\pi}$ and $Z = 2\sqrt{6} \sqrt{\pi}/(3\kappa)$, so at $\kappa = \frac{1}{2}$ one has $4/Z^2 = 3/8\pi$ and $Z = \sqrt{32\pi/3}$. This **scale–fraction split** is the structural reason a forcing of κ must come, if at all, from a condition on the *action normalization*, not the spectrum. We test the three candidate conditions.

3. The unforceability theorem

We pose κ on the framework's covariant modified-inertia matter action (a Galley doubled-field, ϕ_- -linear, non-local in-in functional whose variation sources zero metric — genuinely modified inertia, Cassini-safe by class), and ask whether its overall normalization is fixed. All three standard conditions fail, each for a distinct, machine-verified reason.

(i) Ghost-freedom is blind to κ — because κ is an overall normalization. κ multiplies a degree-1-homogeneous (ϕ_- -linear) action; the equation of motion $N m(\ddot{q}_+ + \omega^2 q_+) = 0$ divides cleanly by the normalization N , so the trajectory is identical for every $N > 0$. Every no-ghost / gradient-stability inequality and every sound speed $c_s^2 = B/A$ is N -invariant; the Einstein-aether analog has all spin-2/1/0 speeds depending only on dimensionless coefficients, with $1/16\pi G$ cancelling. The modified-inertia gate is healthy for *every* $a_0 > 0$ (no interior ghost boundary singles out a value). Ghost-freedom cannot fix even the scale, let alone κ .

(ii) Unitarity reaches only signs, causality, and ratios — never an absolute normalization. Schwinger–Keldysh / in-in positivity yields phase conditions (N -invariant), sign-of-coupling conditions, and noise/dissipation *ratios* with a temperature, but no absolute dimensionful scale. For the conservative (even) kernel the retarded part has zero imaginary part, so the fluctuation–dissipation relation is vacuous. Unitarity is scale-unreachable across the board.

(iii) Holography reaches only the scale cH — never the fraction. From $\{G, \hbar, c, H\}$ the *unique* $\hbar$ -free acceleration is cH (the monomial family $a \sim G^b \hbar^b c^{1-5b} H^{1+2b}$ is $\hbar$ -free only at $b = 0$). The sole dimensionless group, $G\hbar H^2/c^5 \sim 1/S_{\text{dS}} \sim 10^{-122}$, is $\hbar$ -laden and cancels from any classical a_0 . CKN/Bousso/de Sitter-equipartition are *inequalities* saturated with a free $\mathcal{O}(1)$; fixing that $\mathcal{O}(1)$ to the target is *choosing* κ , which is circular.

The needed number $1/Z = \sqrt{3/32\pi} = 0.1727$ is built from the Friedmann 3, the Einstein 8π , and the free-fall $\frac{1}{2}$ — none of which a sign, a ratio, or an entropy can see. Hence $\kappa = \frac{1}{2}$ **is provably unforceable** by healthiness, unitarity, and holography.

4. The Standard-Model degree-of-freedom count does not fix it either

The one remaining principled avenue is to fix the action normalization by a *microscopic degree-of-freedom count* — and the CKN UV–IR bound is exactly that physics. In an IR box of size $L = 1/H$, an effective field theory with cutoff Λ must not collapse to a black hole: $\rho L^3 \lesssim (L/2)M_P^2$, saturated at $\Lambda_{\max} \sim \sqrt{M_P H}$, and the framework sits precisely at this saturation, $\rho_{\text{DE}}^{1/4} = \sqrt{2/Z} \sqrt{E_P E_H}$, with $\sqrt{2/Z}$ in CKN’s free $\mathcal{O}(1)$ slot.

We computed the CKN coefficient *independently* from the honest Standard-Model count, $g_* = 106.75 = 28_{\text{boson}} + \frac{7}{8} \cdot 90_{\text{fermion}}$, with the value $\sqrt{2/Z}$ quarantined out of every solve. The result is decisive (Table 1): the framework coefficient $\sqrt{2/Z} = (3/8\pi)^{1/4} = 0.58779$ is **pure geometry** — sympy confirms it carries no free symbols, no g_* , no thermal factor — while the honest CKN coefficient is an explicit *function* of g_* (energy version $\propto g_*^{-1/4}$, entropy version $\propto g_*^{-1/3}$). A constant and a strictly decreasing function of g_* coincide at exactly one value of g_* ; demanding agreement is *solving for g_** , and the roots ($g_* \approx 3\text{--}25$) are nowhere near the Standard-Model 106.75. At the real g_* the CKN coefficient is $\sim 0.18\text{--}0.41$ (energy) and $\sim 0.31\text{--}0.41$ (entropy) — never within 5% of the framework value.

	framework $\sqrt{2/Z}$	CKN, honest SM ($g_*=106.75$)	CKN, $g_*=1$ limit
coefficient	0.58779	0.18–0.41	0.58779
g_* -dependence	none (geometry)	$g_*^{-1/4}$ (energy)	—
matches framework?	—	no ($\sim 0.4\text{--}0.7\times$)	yes (identity)

Table 1. The framework’s $\mathcal{O}(1)$ versus the honest CKN coefficient.

The geometric origin. The exact landing occurs only in the $g_* = 1$, no-thermal, ball-volume $(4\pi/3)$, half-Schwarzschild ($\frac{1}{2}$) convention — and sympy proves to an *exact zero* that this is the geometric identity

$$\frac{3}{8\pi} = \frac{(\text{Schwarzschild } \frac{1}{2})}{(\text{ball } \frac{4\pi}{3})}.$$

So the framework’s coefficient is **the single-degree-of-freedom limit of the CKN bound** — CKN saturation with the matter count set to one. This is precisely *not* a microscopic fix: the value lives where the Standard-Model count has been discarded. κ is geometry, not particle content, and this closes the last named avenue.

5. What κ is

Across four independent routes — ghost-freedom, unitarity, holography, and a Standard-Model degree-of-freedom count — $\kappa = \frac{1}{2}$ cannot be forced, and always for the same underlying reason: **its value is pure geometry**, the Einstein 8π and the Friedmann 3, fixed by the field equations before any matter, EFT, spectrum, or dof count is specified. It is the residual fraction outside the gravitational root, equivalently the $g_* = 1$ geometric limit of the holographic bound. A sign, a ratio, an entropy, and a particle count are each structurally blind to it.

6. What this does and does not establish — both ways

- **Credit.** The framework is a *provably* one-parameter theory of gravity and the dark sector: $\kappa = \frac{1}{2}$ the lone free number, with the form $a_0 \propto c^2 \sqrt{\Lambda}$ and the kernel $\sqrt{8\pi/3}$ (including $\sqrt{\pi}$) forced. One knob tying the MOND scale to Λ , no dark-matter particle — maximally economical against Λ CDM’s six parameters plus an undetected particle. We now know *exactly why* the one number cannot be derived (it is geometry, carrying no microscopic content) and *what it is* (the single-dof limit of CKN). The unforceability is established non-circularly: κ was kept symbolic throughout; the value $\frac{1}{2}$ and the 8π density route enter only the post-hoc locating map, and the $g_* = 1$ identity *fell out* of an independent derivation rather than being assumed.
- **Concede.** a_0 ’s *value* is **not** derived from Λ ; the sector is one-parameter, **not** zero-parameter. The de Sitter–Unruh temperature route’s natural scale is $2cH_\Lambda$ ($\sim 12\times$ too large); the empirical a_0 requires adopting the density route plus the $\kappa = \frac{1}{2}$ posit, and no condition forces that choice. The full joined matter+metric ghost analysis is represented, not completely closed. And — most importantly — this is a **structural / consistency theorem about the framework’s own parameter economy; it does not by itself confirm the framework against Λ CDM**. A theory can be maximally economical and still wrong; that verdict belongs to the data.

7. Closing

It is unusual to be able to *prove* that a free parameter is irreducible. For the de Sitter–MOND acceleration scale we can: the lone coefficient $\kappa = \frac{1}{2}$ resists ghost-freedom, unitarity, holography, and a Standard-Model degree-of-freedom count, because its value is pure geometry — the single-degree-of-freedom limit of the Cohen–Kaplan–Nelson bound, the Einstein 8π over the Friedmann 3. The framework is therefore a one-parameter theory of gravity and the dark sector, its single knob characterized exactly and shown to be unforceable. That is the honest summit of the structural program: not a derivation of a_0 , not a theory of everything yet, as frustrating as it may be — but a sharp statement of *how few* numbers the proposal costs, and *why* it cannot cost fewer.

Companion to the de Sitter–MOND framework (DOI [10.5281/zenodo.20721540](https://doi.org/10.5281/zenodo.20721540)) and the falsifiable $a_0(z)$ prediction (DOI [10.5281/zenodo.20737162](https://doi.org/10.5281/zenodo.20737162)). All identities — $\sqrt{2/Z} = (3/8\pi)^{1/4}$, $3/8\pi = (\frac{1}{2})/(\frac{4\pi}{3})$, $\kappa = \frac{1}{2} \Leftrightarrow a_0 = c^2 \sqrt{\Lambda/32\pi}$, and the g_ -scaling of the CKN coefficient — are machine-verified and reproducible from the repository (github.com/carlzimmerman/zimmerman-formula, [opus_48_extended_research/](#)). The open questions — the un-derived value of a_0 , the phenomenological covariant lensing sector, and the absence of a Standard-Model derivation — are real and stated in the technical record. This is a structural result for a falsifiable theory of gravity and the dark sector; it is not a proven replacement for dark matter, and not a theory of everything.*